Deadline: Feb. 22nd, 2022.

1.6(a) A perfect gas undergoes isothermal compression, which reduces its volume by 2.20L. The final pressure and volume of the gas are 3.78×10^3 Torr and 4.56L, respectively. Calculate the original pressure of the gas in (a) Torr, (b)atm.

1.13(b) A gas mixture consists of 320 mg of methane, 175 mg of argon, and 225 mg of neon. The partial pressure of neon at 300 K is 8.87 kPa. Calculate (a) the volume and (b) the total pressure of the mixture.

1.16(a) Calculate the pressure exerted by 1.0 mol H_2S behaving as (a) a perfect gas, (b) a van der Waals gas when it is confined under the following conditions: (i) at 273.15 K in 22.414 L, (ii) at 500 K in 150 cm^3. (a=4.484 atm L^2 mol^{-2}; $b=4.34 \times 10^{-2}$ L mol^{-1}).

1.18(b) A gas at 350 K and 12 atm has a molar volume 12 percent larger than that calculated from the perfect gas law. Calculate (a) the compression factor under these conditions and (b) the molar volume of the gas. Which are dominating in the sample, the attractive or the repulsive forces?

1.19(a) In an industrial process, nitrogen is heated to 500 K at a constant volume of 1.000 m^3. The gas enters the container at 300 K and 100 atm. The mass of the gas is 92.4 kg. Use the van der Waals equation to determine the approximate pressure of the gas at its working temperature of 500 K. For nitrogen, a = 1.352 L^2 atm mol^{-2}, b = 0.0387 Lmol^{-1}.